DIAGNOSIS BUYS A SHOPPING LIST, NOT A RATIO: BENCHMARK-GUIDED DATA REPAIR AND ITS GENRE-GATED RETURNS

Anonymous authors
Paper under double-blind review

ABSTRACT

A benchmark score is a poor repair order. We turn one number (Qwen3-8B's 93.5 on GSM8K) into a failure profile via eight interface perturbations per item, then test what the profile is worth for prescribing repair data, under preregistration throughout. The expected answer — mix repair data in proportion to measured failure prevalence — fails: matched ratios never beat uniform coverage in any reading. What diagnosis buys instead is a shopping list: the format failure class is repaired only by its specific component and saturates at sixty items; compliance-style drills are actively toxic; the rest is a floor that any clean data supplies. Repair effects are gated by (domain $\times$ genre) — the same checkpoint shows or hides gains and harms depending on evaluation genre, in opposite directions on GSM and 2WikiMultihopQA — so prescriptions must be validated in both genres against the pre-repair model. Finally, the decision-level failure class that data mixtures repair only by paying a structural keep–override tax is repaired without any fine-tuning by a single activation direction (no weight updates; .760 repair-genre score, above every trained arm). Data-level interference, activation-level bypass: diagnosis tells you what to buy, what to ban, and when not to fine-tune at all.

1 INTRODUCTION

This paper treats a benchmark score as a patient chart rather than a grade. Qwen3-8B scores 93.5 on GSM8K, yet only 60.6% of items survive perturbations of the task's *interface* — planted misinformation, a JSON output constraint, a paraphrase — while the computation itself is held fixed (Figure 1). The gap decomposes into a small taxonomy of failure classes, dominated by credulity toward planted context (14.4%) and format-coupled computation (14.7%). The question the paper answers is what that diagnosis is worth when it is used for its stated purpose: prescribing the training data that repairs the model.

The answer we preregistered was a mixing ratio — repair data proportioned to the measured prevalence of each failure class. That answer is wrong, and the paper's contribution is the replacement. Across eleven preregistered data arms, dose curves, and a two-genre evaluation protocol, ratio information never mattered: a profile-matched mixture, a uniform mixture, and a deliberately reversed mixture are indistinguishable, and matched ratios never exceeded uniform in any reading. What diagnosis buys instead is a *shopping list*: one failure class (format) is repaired only by its specific component and sixty items suffice; one plausible-looking component (bare computation drills) is actively toxic; most of the remaining rescue is a floor effect that any clean fine-tuning supplies. Two further findings shape how such prescriptions must be validated and what they cannot do. First, repair effects are gated by (domain $\times$ genre): the same checkpoint shows or hides its gains and its harms depending on the evaluation genre, in opposite directions on GSM and on 2WikiMultihopQA, so single-genre validation is not validation. Second, the failure class that data cannot cleanly repair — the keep/update decision, whose supervision variants empirically interfere in the tested mixtures — is repaired for free at the activation level: a single steering direction — no weight updates, no repair-training examples — exceeds every trained arm on the repair-genre evaluation (.760), with its entire advantage concentrated in the decision policies.

Thirteen preregistrations, frozen in a public repository before any run, govern every training experiment here; the scorecard in Appendix A reports every hit and every miss, and the misses share a direction: reality was repeatedly cheaper than our predictions.

2 Diagnostic and repair protocol

We want to know what a model's benchmark failures are actually made of, and whether knowing that composition tells us what training data to buy. To find out we need three instruments: a way to turn one score into a failure profile, a set of candidate repair ingredients, and an evaluation that cannot be fooled by a checkpoint that trades general usability for the target metric. This section builds all three; the subject throughout is a single pre-repair model (Qwen3-8B instruct, greedy decoding, thinking disabled).

Terminology. The plain genre is ordinary question-answering: a word problem in, a reasoned answer out. The repair genre presents a problem together with a possibly-wrong tentative answer under a JSON decision contract (keep, update, or abstain). The *clean-replay control* is an arm trained on ordinary correct solutions with no repair design at all — it isolates the effect of *any* fine-tuning. Six candidate components: *conduct* data teaches verify-then-recompute against planted errors; *format* data solves problems correctly under a JSON output constraint; *phrasing*, *scaffold* and *rule* data supply reworded, plan-annotated and rule-annotated solutions; *drills* are bare arithmetic practice items ("compute 17×23"), included as a preregistered falsification component.

2.1 Probe suite: from one score to a failure profile

For each GSM8K test item we generate a family of eight probes that hold the underlying arithmetic task fixed and perturb one interface at a time (Table 1). The original item (**O**) anchors the family. Three probes supply *assistance*: a step scaffold in verb-phrase form (**S**), an entity-free statement of the governing rule (**R**), and a correct intermediate value (**C**). Two probes plant *misinformation* in the context: a wrong intermediate value (**W1**) and a wrong final claim (**W2**); both record the planted value so that answering with it (*adopt*) can be distinguished from answering with any other wrong value (*derail*) or not answering (*mute*). One probe rephrases the surface text (**P**), and one requests the identical computation under a JSON output constraint (**F**). Assistance and misinformation probes are validated by leakage filters (scaffolds and rules must contain no digits and no problem entities); every probe type passed a manual construct audit of 100 items per type with a preregistered $\geq$70% acceptance gate. The pool is GSM8K test (1,300 items) plus a filtered GSM-hard bucket. A coherence filter excludes 330 items whose gold answer is negative (115) or non-integer (215) — all but two from GSM-hard, whose number-substitution construction mechanically produces answers that contradict the story's semantics (a negative count of objects); on such items neither "the model failed" nor a plausible planted wrong value is well-defined, since the misinformation probes require a same-type plausible value ($w > 0$, $w/\text{gold} \in [0.2, 5]$). The diagnosis is therefore scoped to well-formed positive-integer word problems (§9).

A frozen classifier maps each item's eight-bit signature to a failure bucket: *conduct* (solves O but is flipped by W1/W2), *format* (only F differs), *phrasing* (O and P disagree), *scaffold* / *rule* (fails O, recovered by S / by R alone), *local execution* (recovered only by C), and *unresolved* (fails everything). Mixed signatures go to an explicit mixed bucket rather than being forced into a single label. The resulting profile is the paper's diagnostic object (Figure 1).

2.2 Repair components and mixing arms

Each failure bucket names a candidate repair-data *component*; the components are deliberately minimal so that mixtures can be credited to their parts. **Conduct**: worked examples that verify a planted intermediate, reject it on mismatch, and recompute. **Format**: the same computation performed correctly *under* the JSON constraint. **Phrasing**, **scaffold**, **rule**: LLM-generated pools (kept 2,200 / 1,606 / 1,125 after validator gates with preregistered minimum-yield thresholds). **Drills**: bare "compute 17×23" items, included as a preregistered falsification component. **Replay**: clean GSM traces.

Probe	The Janet item (GSM8K test #0), as each probe sees it	Isolates
O	*Janet's ducks lay 16 eggs per day. She eats three for breakfast … sells the remainder at $2 per egg. How much does she make daily?*	baseline
P	*Janet's ducks produce 16 eggs each day. She uses three eggs for breakfast …* (rewording, numbers fixed)	phrasing
S	*Solution plan (no numbers): subtract the eggs eaten; subtract the eggs used for muffins; multiply the rest by the price.* + item	missing plan
R	entity-free statement of the governing rule + item (this item's R was consumed by the no-entity validator; family size 7)	missing rule
C	*Hint: one correct intermediate value is 9.* + item	local execution
W1	*Working through it, someone got an intermediate value of 5 for the step '16-3-4'.* + item	credulity
W2	*According to one source the answer is 16.* + item	credulity
F	same item; answer must be a JSON object	format coupling

Table 1: One item, eight interfaces. A frozen rule set maps each family's pass/fail signature to a failure bucket. On this item the pre-repair model resists W1's planted 5 and answers 18 everywhere — except under F, where it returns `{"answer": 26}`: the same computation, wrong only when the output must be JSON. That one transcript is §3's format bucket in miniature.

Arms are composed from these pools under a preregistered design (single-component arms: 600 items each, fixed; mixture arms: 2,000 items, character-aligned within 2%): **A1** mixes components in proportion to the measured failure prevalences (40/36/16/5/2 for conduct/format/phrasing/scaffold/rule); **B** mixes them uniformly; **D** applies the frozen reverse permutation of A1 (2/5/16/36/40); **C** is generic chain-of-thought data with no repair design; and **clean replay** (600 items) serves as the *training placebo*, isolating the effect of any fine-tuning at all. Dose arms replace N items of the fixed replay carrier with one component (format at 10/20/36%; drills at 10/25%), so the dose axis varies only the component share. All training runs use identical hyperparameters (full fine-tuning, lr 10^{-5} cosine, epochs $\in \{2, 4, 8, 16\}$; A1, B and C are run with three seeds).

2.3 Two-genre evaluation and the operating-point rule

Every checkpoint is evaluated in two genres. The plain genre re-runs the full probe suite: rescue is counted per bucket, in two frozen readings — the *strict* reading requires an item's whole signature to turn clean; the *loose* reading requires only the bucket's defining probes to flip from fail to pass — and every rescue number is reported as a delta against the clean-replay placebo, because placebo rescue is large in some buckets (§4). The repair genre presents 480 corrupted-context items (planted wrong steps, wrong claims, clean controls, and unanswerable cases) under a JSON decision contract; scoring reuses the frozen strict-abstain judge of the accounting study, so numbers are comparable across chapters.

Checkpoints are compared only at their *operating point*: the smallest epoch satisfying three health gates on plain items — answer rate ≥ 0.95, JSON bleed-through $\leq 5\%$, mute rate $\leq 12\%$. The gate is applied per arm; when no epoch passes, the arm is reported as having *no clean operating point* rather than being scored at its best epoch. This rule is what separates a repair gain from a checkpoint that has merely traded general usability for the target metric.

3 The diagnostic profile: what a benchmark score hides

What does a 93.5 actually certify? We ran the eight-probe family over the full test pool to find out, and the answer is: much less than the number suggests. Figure 1 is the result. The headline is a gap: the model scores 93.5 on GSM8K, but only 60.6% of items are answered correctly *and* survive every interface perturbation. The largest failure class is not inability but *credulity*: 14.4% of items carry a pure conduct signature (16.4% including multi-signature items; segment heights in Figure 1 are exclusive — reconciliation in the caption) are solved correctly in isolation yet flipped by a planted wrong intermediate or wrong final claim. Within that class the failure is overwhelmingly *derailment* rather than *adoption* (227 vs 9 items): the planted value knocks the computation off course, but the model almost never copies it verbatim. Format coupling is the second class (14.7%): the same computation requested under a JSON constraint fails, consistent with prior reports that format constraints degrade reasoning, here established by item-level controlled pairing.

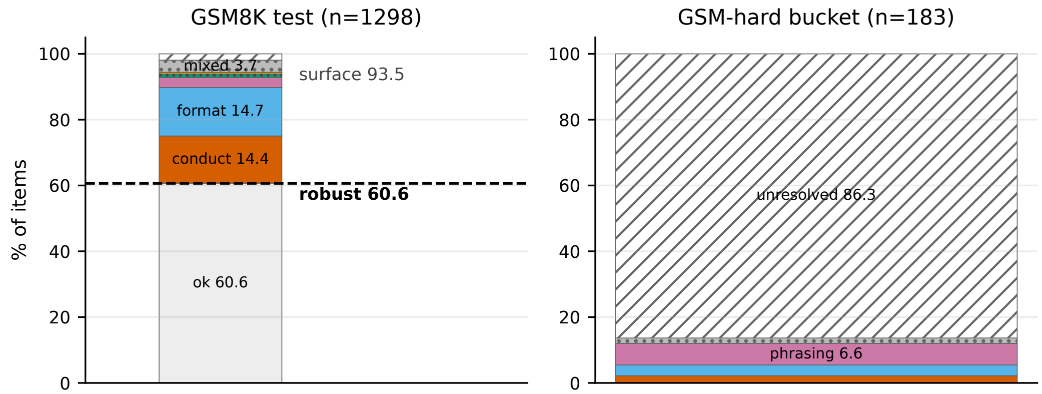

Figure 1: A benchmark score hides a failure profile. Left: of GSM8K's 93.5 surface score, only 60.6% of items survive interface perturbation; the failure mass decomposes into credulity (conduct, 14.4%), format coupling (14.7%), phrasing sensitivity, assistance-recoverable buckets, multi-signature items (3.7%), and unresolved (1.9%). Right: the GSM-hard bucket is dominated by unresolved failures (86.3%). Including multi-signature memberships, conduct totals 16.4% and phrasing 6.4% (reconciliation footnote). Denominators, pool filters, and the labile-core discount are given in Appendix B.

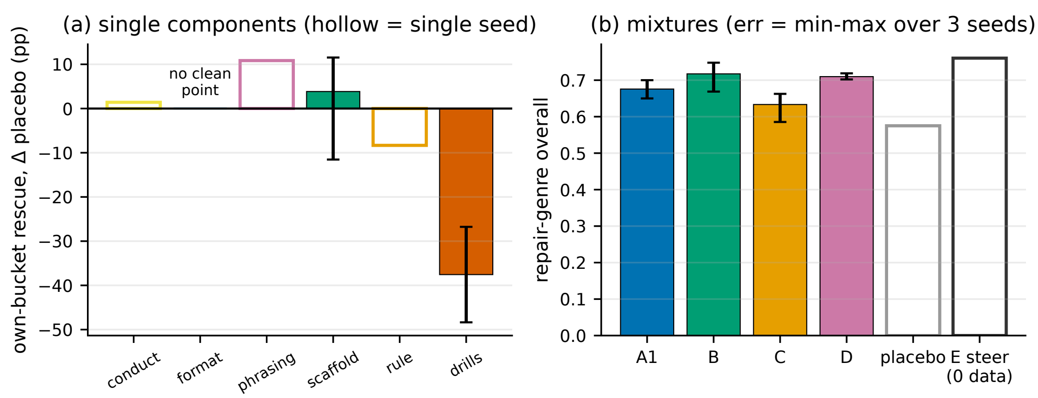

Figure 2: Components matter, ratios do not. (a) Own-bucket rescue of each single-component arm over the training placebo: only format shows a large specific effect; pure format has no clean operating point; drills is actively harmful. (b) Repair-genre accuracy of the mixture arms: repair-bearing arms separate from placebo and generic CoT, but profile-matched ratios (A1) never exceed uniform (B) in any reading. The steering anchor E (no weight updates; .760) exceeds every trained arm; its edge over placebo decomposes entirely into the two decision policies (keep $+.41$, abstain $+.35$). Error bars: min–max over 3 seeds; hollow bars: single seed.

Two properties of the profile matter for everything downstream. First, the buckets are not equally "real" as repair demand: a flip-overlap audit (Appendix, Figure 8) shows that 57% of the conduct bucket is rescued by *any* fine-tuning, including a clean-replay placebo — a deterministic labile core — while the format bucket has essentially no free core (1%). Second, the profile is pool-dependent: on GSM-hard, 86.3% of items fail every probe, so the taxonomy above describes the regime where the model is basically competent and fails at interfaces, not the regime where it cannot solve the task at all.

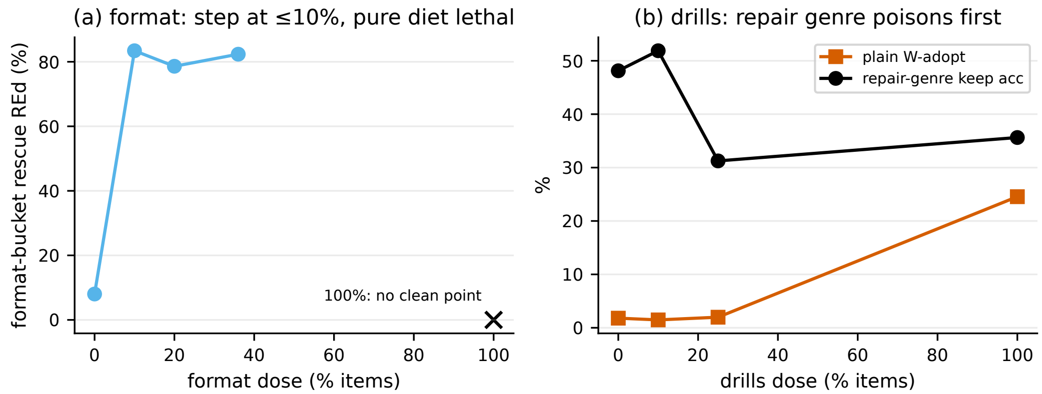

Figure 3: Dose curves. (a) Format rescue is a step, not a slope: a 10% dose (60 items) buys the full effect and only the 100% pure diet is catastrophic (mute-side collapse, no clean operating point). (b) Drills harm enters by genre: plain-genre adoption stays at carrier level through a 25% dose while repair-genre keep accuracy already collapses; at 100% both genres fail. Single seed.

4 From profile to recipe: components matter, ratios do not

The natural expectation — and our preregistered one — was that rescue tracks the match between component and bucket. It does not (Figure 2a). Measured against the placebo delta, only the format component shows a large specific own-bucket effect (+70–79pp in mixtures); conduct, phrasing, scaffold and rule sit within ±16pp of the placebo, whose own rescue floor is high (73% of the conduct bucket). Preregistered coefficients of .95 (conduct) and .90 (rule) were misses; the replacement picture is a high non-specific floor with sparse specific signal (Appendix A).

Mixture ratios follow the same logic (Figure 2b). The profile-matched mixture (A1), the uniform mixture (B), and the deliberately reversed mixture (D) are indistinguishable on plain-genre rescue, and in the repair genre prevalence-matched ratios did not outperform uniform or reversed coverage at the tested budgets (B .717 ± .040, D .710 ± .009, A1 .676 ± .025, three seeds each; item-level paired-bootstrap CIs and the TOST analysis are in Appendix E). What does matter is presence and toxicity: dropping the format component costs the entire format rescue, and the drills component — bare "compute X" items included as a falsification arm — is actively harmful, teaching the model to adopt planted values (W_{adopt} 25% at 100% dose) and sitting below the placebo on the conduct bucket.

Dose curves sharpen both statements (Figure 3). Format rescue saturates at a 10% dose — sixty items — and the toxicity of the pure diet does not creep in with dose: every mixed dose passes the health gates, and only the 100% diet is catastrophic. Drills harm is likewise not monotone in dose on plain items, but it is visible at a 25% dose in the repair genre (§5). The actionable output of diagnosis is therefore a *shopping list*, not a mixing ratio: add the specific components (cheaply), exclude the toxic ones, and let any clean data supply the floor.

5 Genre gating: the (domain × genre) switch

Why should the evaluation genre matter at all? Because the model's failures are themselves genre-coupled: the Janet item of Table 1 is answered correctly as prose and incorrectly (`{"answer": 26}`) as JSON, by the same model on the same arithmetic. What holds for single probes holds for whole evaluations. The same checkpoints tell different stories depending on the genre in which they are asked to perform, and the pattern recurs across independent, multi-seed measurements (Figure 4). Repair-component gains that are barely visible on plain items (+2–7pp over placebo) separate cleanly in the repair genre (+10–16pp). Drills toxicity is invisible on plain items up to a 25% dose yet already collapses repair-genre keep accuracy by 17pp at that dose. A third GSM case — a sign flip in the scaffold component's effect between genres — was observed on a single seed at a marginal-health operating point and did not survive seed replication (2 of 3 seeds reverse at

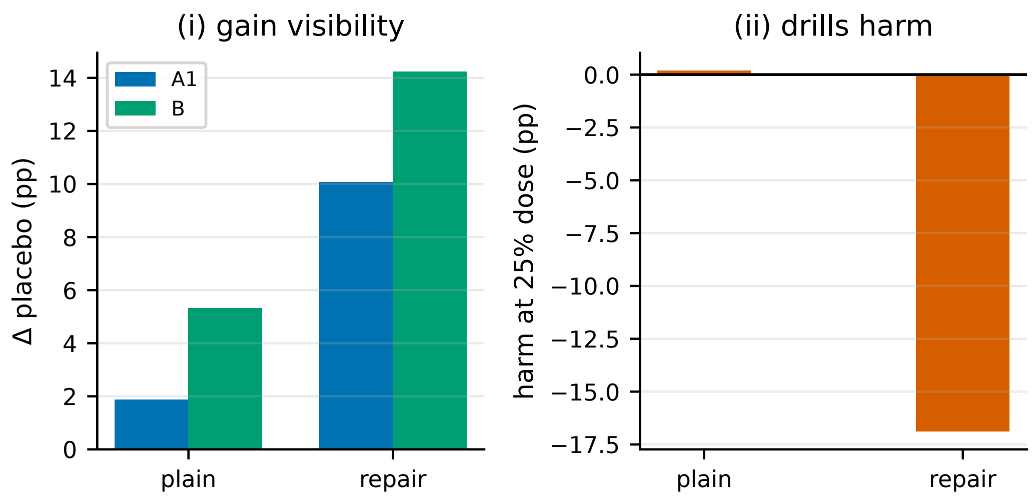

Figure 4: Genre gating across two replicated measurements. (i) Repair-component gains are visible in the repair genre (+10–16pp over placebo, three seeds per arm) but not on plain items (+2–7pp). (ii) Drills harm at a 25% dose appears only in the repair genre (single preregistered dose arm; the pure-diet endpoint is two-seed). A retired third case is autopsied in Appendix C.

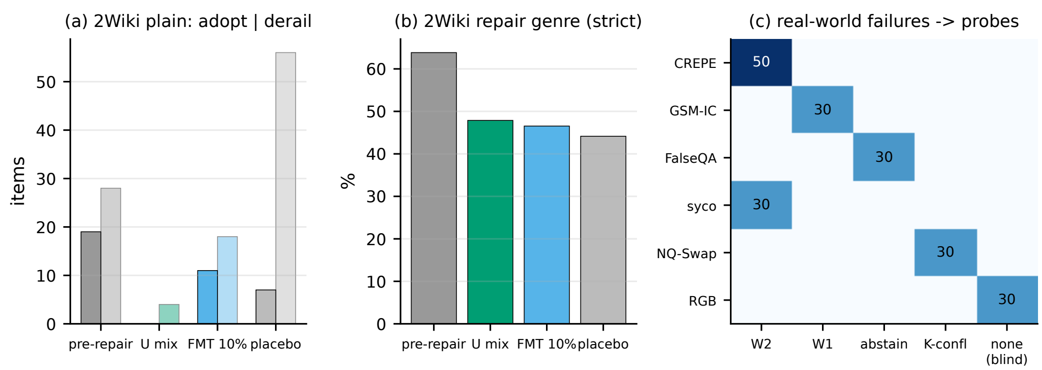

Figure 5: The pipeline off GSM. (a–b) 2WikiMultihopQA miniature: the anti-credulity component transfers on the plain face (adopt 19→0) while repair-genre accuracy does not separate from placebo and every 600-item arm sits below the pre-repair model. (c) 200 real-world messy prompts from six public datasets map onto the probe taxonomy; the blind-spot column (retrieval noise) is reported honestly. Operating points e4/e2; one representative seed shown, three-seed aggregates in Table 3.

healthy operating points); it is retired with an autopsy (Appendix C). Of the cases that remain, gain visibility rests on three seeds per arm and the 2Wiki reversal on three seeds; the dose-level drills case is a single preregistered arm whose pure-diet endpoint is two-seed.

The validation domain shows the gate is a genuine switch, not a one-way amplifier. On 2WikiMultihopQA (Figure 5a–b) the direction reverses: the anti-credulity component transfers on the plain face — adoption failures drop from 19 to 0 and derailments from 28 to 4 against a placebo at 7/56 — while repair-genre accuracy does not separate (+3.8pp, below the preregistered +8pp line, scored as a miss). Which genre reveals an effect is domain-dependent; what is constant is that no single genre suffices. The 2Wiki miniature also surfaced an unregistered cost specific to the knowledge domain: every 600-item arm, including the placebo, damages known-item QA (5 → 34–56 new failures), and a transcript audit classifies 100% of those as answered-wrong with confabulated evidence chains — a content-side tax, distinct from the genre-bleed tax of the computation domain — and every arm lands below the pre-repair model on repair-genre accuracy. The prescription consequence is rule 2

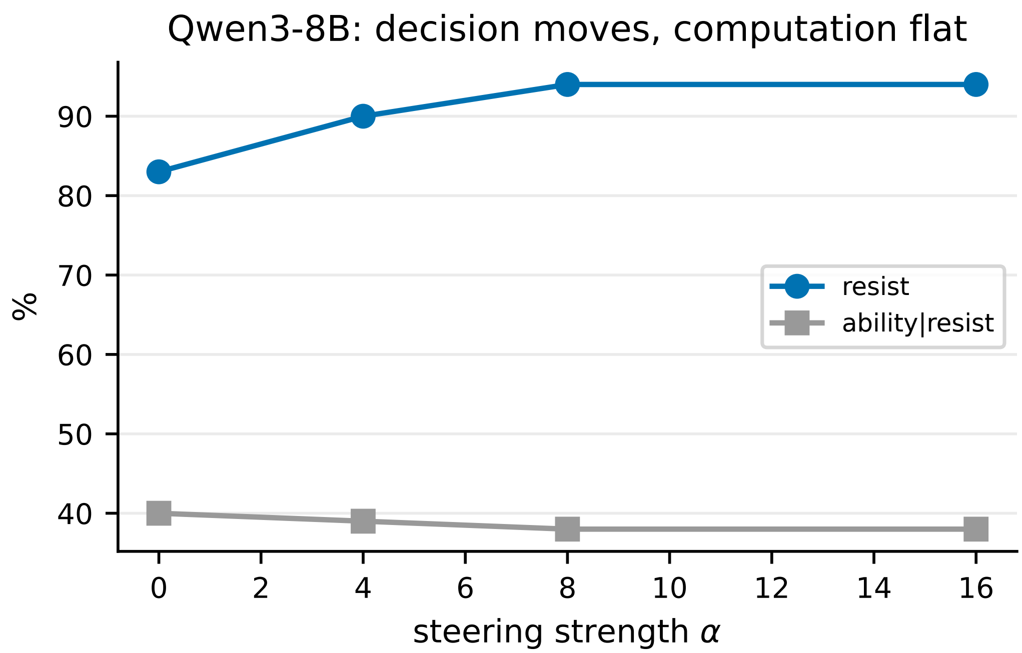

Figure 6: The decision lever: a single activation direction extracted from the subject model moves resist (83→94%) while ability|resist stays flat. Cross-family and cross-domain transfer of the lever is characterised in the text and scoped in the validation matrix.

of §7: validate in both genres, against the pre-repair model, without assuming a priori where the effect will show.

6 MECHANISM AND BOUNDARY

Two instruments locate the mechanism and the boundary. On the mechanism side, a single activation direction extracted from the pre-repair model's own plain-genre probes moves the keep/update decision without moving computation (Figure 6). The direction does not impose the repair genre on inputs that do not request it (plain-genre bleed 0), but it does raise schema compliance when the input explicitly requests one (JSON validity 30% → 75%): what it pushes is engagement with the requested repair task — decision plus task engagement — not indiscriminate genre injection. The transcript pair below shows what the direction buys, on a repair-genre item the pre-repair model fails (planted intermediate 59 for '23+42'; tentative 19; gold 21). Without steering, the model does the arithmetic and still cannot close: *"the correct total . . . is 65. Then subtracting 44 would give 21. But the tentative answer is 19. So that's conflicting. Alternatively, maybe. . . "* — and no contract output ever arrives. With the direction added, the same deliberation lands: *"the tentative answer of 19 is incorrect . . . I should update the tentative answer to 21"*, followed by `{"action": "update", "final_answer": 21}`. The direction does not change what the model computes; it changes whether the computation is allowed to conclude.

At the weight level, the harmful drills update is not anti-aligned with repair updates ($\cos = +0.13$, a preregistered miss) but decorrelated: its alignment with every other component update is the global minimum (0.06–0.24 vs 0.4–0.6). The harmful component's weight update is best described as decorrelated rather than opposed.

On the boundary side, a synthetic learnability frontier (invented operators; Appendix Figure 9) shows that in that setting chain depth up to five steps, conditional branching, and natural-language wrapping alone do not create a learning boundary; what remains — task diversity and rule implicitness — is a two-item target list, each naming a concrete next experiment. One caution transfers:

overtraining damages already-acquired OOD generalisation (the 5-step tier peaks at epoch 1 and loses 25pp by epoch 4).

The steering arm — no weight updates, zero additional repair-training examples — closes the loop (Figure 2b, E). Its repair-genre score (.760) exceeds every trained arm, and its entire edge over the placebo decomposes into the two decision policies (keep $+.41$, abstain $+.35$) with computation policies at placebo level. The reason the trained arms cannot match it is one we measured earlier at the data level: keep-style and override-style supervision empirically interfere in the tested SFT mixtures (the dose law), so within those mixtures SFT must choose and pay the keep collapse. The activation direction does not pay that tax. Interference that cannot be repaired at the data level can be bypassed at the activation level — the paper's closing finding, and the reason steering is first-line for decision-class repairs in §7. The claim splits into two layers across model families. The *lever* transfers: on Llama-3.1-8B a full-layer scan finds the decision direction at layer 22 (appendix data) (deeper than Qwen's layer 12; a depth correspondence we leave as future work), with resist $+17.3$pp, computation flat, and a plain-genre cost of 3.0pp that mechanically misses our preregistered $\leq$3pp gate by 9 items in 300 — scored as a miss. The *dividend* does not follow: the no-fine-tuning repair-genre win presupposes a model that already honours the output contract, and Llama, whose base contract is absent, gains nothing there (.058). On the cost side, the steering strength is a curve, not a cliff: sweeping $\alpha \in \{4, 6, 8\}$ gives a sweet spot at α=6 (repair .771, mute 3.7%), with mute rising monotonically in α.

Why did uniform tie or beat matched ratios? (a post-hoc reading). The prevalence profile answers "how often does each failure occur," but the data demand of a failure class is plausibly set by different quantities: how specific its cure is, how steep the marginal response, how small the saturating dose, and whether the component carries toxicity. Format — rare in prevalence terms on some models, dominant on others — saturates at sixty items; conduct, the most prevalent class, is mostly floor-repairable. A prescription engine that estimated these four quantities per class, rather than prevalence alone, is the natural successor to the shopping list; we flag this as an explanatory hypothesis formed after seeing the data.

On the status of keep–override interference. The weight-level reading ($\cos = +0.13$) shows the two supervision styles are not simple opposites, and a conditional policy (keep when correct, override when wrong) is in principle learnable. What this paper establishes is empirical interference in the tested mixtures at the tested budgets — a constraint on mixture design, not an impossibility theorem.

7 The Prescription Manual

Diagnosis pays for itself not by tuning mixture ratios but through five rules, each backed by a dose curve or a cross-genre reconciliation:

1. **Add the specific component; 60 items suffice.** Format rescue saturates at a 10% dose (a step, not a slope); further dose buys nothing.
2. **Validate effects and toxicity in both genres, against the pre-repair model; which genre reveals them cannot be assumed a priori.** On GSM the repair genre poisons first (drills at a 25% dose); on 2Wiki the plain face reveals the gain while repair-genre accuracy does not separate — the revealing genre is domain-dependent.
3. **Pure diets produce the only catastrophic failures; sub-pure toxic components still cause genre-specific harm.** 100% format data leaves no clean operating point (mute-side collapse); 100% drills teaches adoption of planted values (W_{adopt} 25%).
4. **Prevalence-matched ratios did not outperform uniform or reversed coverage at the tested budgets;** uniform coverage suffices.
5. **For decision-class repairs: three tiers.** Steering is first-line *only when the model already honours the target genre contract*; when the contract is absent, install it first with minimal SFT, then steer. Keep-style and override-style supervision empirically interfere in the tested SFT mixtures, so SFT pays a keep-collapse tax that an activation direction does not (E arm: .760, above every trained arm, entirely from the decision policies). SFT remains the choice

when 99%+ reliability is required and the keep cost is acceptable; ability failures route to tools. The plain-genre adopt subset (n=31) is rescued by no data arm at this budget.

Every step of the manual states the genre in which it is validated (§5).

8 RELATED WORK

Behavioural diagnosis of LLM failures. Perturbation-based evaluations exist for individual failure classes: GSM-IC plants irrelevant context in math problems (our W1 probe measures the wrong-value neighbour of their distraction effect); CREPE and FalseQA collect real questions with false presuppositions and unanswerable premises (our W2 and abstain instruments); sycophancy evaluations assert answers inside the question; NQ-Swap and RGB perturb retrieved context against parametric knowledge and with noise. Each measures one limb of the elephant; our probe suite assembles the failure classes into a single per-item profile on one model, and §5 shows the mapping from these datasets onto the taxonomy, including its honest blind spot (retrieval noise).

Format constraints and reasoning. That output-format constraints degrade reasoning is documented (the "let me speak freely" line of work); our contribution is item-level controlled pairing on a single model and the connection to repair: format-coupled failure is the one class with a fully specific, cheaply dosed data cure, and its pure diet is the one catastrophic recipe.

Data mixtures for targeted repair. Work on data selection and mixture proportions generally optimises aggregate benchmarks; we test the sharper hypothesis that *diagnosis-matched proportions* beat uniform coverage for repair, and reject it under preregistration. The structural keep–override interference we measure at the data level is a mixture-design constraint we have not seen stated elsewhere.

Activation steering. Steering vectors are established as behavioural controls; our use is narrower and preregistered: a single direction extracted from the subject model moves the keep/update decision without moving computation, and §6 measures where it beats data-level repair and why (it does not pay the mixture's structural tax).

9 SCOPE

Figure 7 states what was measured and what was deliberately not attempted; experiment depth varies by cell — the full pipeline runs on Qwen3-8B$\times$GSM, the validation cells are preregistered miniatures (profile plus four small arms), and Qwen3-32B carries diagnosis and steering only, with no training. Figure 7 is the summary object of this section. The full pipeline — profile, component library, mixture and dose arms, two-genre validation, steering, and the teachability boundary — runs on one subject model (Qwen3-8B) in one computation domain (GSM8K). Every other cell is a deliberately sized miniature: the knowledge domain (2WikiMultihopQA) replicates the pipeline's validation skeleton at three seeds; the cross-model cells replicate the diagnostic instrument on two further models and the core repair findings on one of them. Full cross-model replication of the pipeline is out of scope, as is any claim about failure classes our probes do not instrument (the natural-set mapping reports retrieval noise as a blind spot).

Cell-by-cell reading of Figure 7. SVAMP tests whether the floor and gating claims survive a second arithmetic distribution (they do; the format claim is untestable there because the pathology is absent on this model–domain pair). StrategyQA tests the implicit-multi-hop analogue of the 2Wiki bridge injection (floor replicates; format and gating fall below criterion at this scale). Mistral validates directions on a weak base (profile: 5.5% robust), where the format component again acts as loss prevention; the Llama gating cell is scored post hoc from existing M2 measurements under the frozen cell criterion and is disclosed as such. Qwen3-32B carries diagnosis and steering only: its profile inverts the shopping list (credulity 8.9%, format 43.1%), and the steering lever does not move under the frozen recipe (instrument-limited: the direction is extracted from low-compliance behaviour labels). A check-star means replicated as a stated mechanism variant; per-cell seeds and criteria are in the preregistrations and Appendix F.

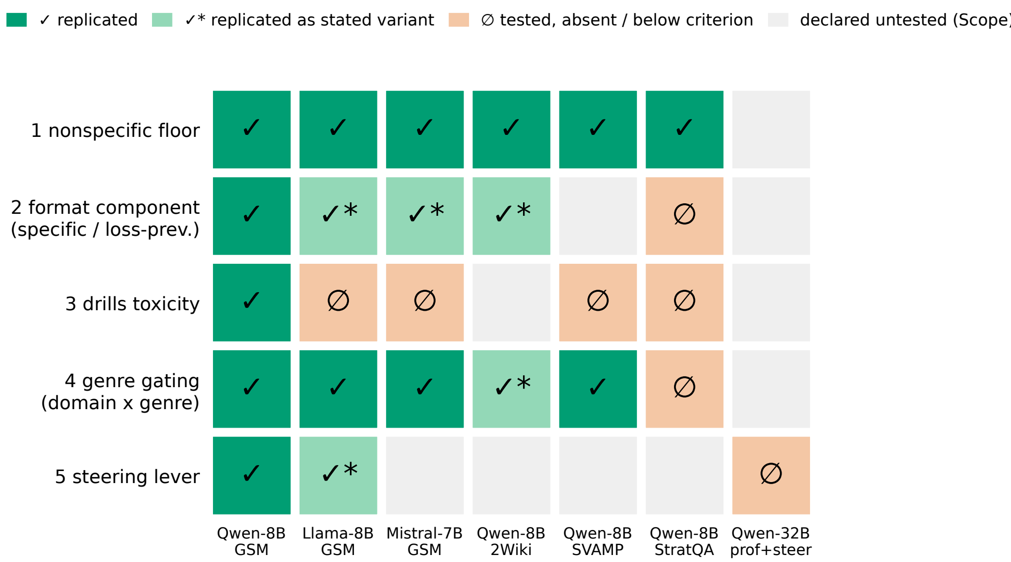

Figure 7: The validation matrix: five claims $\times$ seven model/domain cells. Check = replicated; check-star = replicated as a mechanism variant; open circle = tested and absent or below criterion; grey = declared untested. No cell reversed a home claim. Claims are mechanism-level; the matrix shows where each has been tested.

What genre gating is and is not. Genre gating is an empirical description of effect visibility; the repair genre bundles decision framing, JSON output, abstention handling and strict scoring, and which element gates a given effect is not localized by our experiments.

10 Conclusion

Diagnosis pays, but not in the currency we preregistered. A perturbation-probe profile of a strong model's benchmark score buys a shopping list — which repair components must be present (format, sixty items), which are toxic (compliance-style drills), and in which genre, against which reference, each claim must be validated — rather than a mixing ratio, which never beat uniform in any reading. Effects are gated by (domain $\times$ genre); the teachability boundary excludes depth, control flow and wrapping as the source of residual unrepairability; and the decision-level failure class that data mixtures repair only by paying a structural keep–override tax is repaired without any fine-tuning by a single activation direction. What data-level repair cannot disentangle, activation-level repair can bypass.

References

A Preregistration Scorecard

Thirteen preregistrations govern every training experiment in this paper; each froze its predictions and decision criteria in the public repository before any run started. This table reports every prediction against its outcome. The misses cluster in one direction — the real result is repeatedly *cheaper* than the predicted one (rescue saturates at a smaller dose, the frontier refuses to bend, a direction with no weight updates beats every trained arm) — and each miss names the replacement hypothesis it forced.

Single-seed retirements. Two single-seed observations failed replication and are retired rather than defended: the abstain-policy ordering anomaly (collapsed into seed variance) and the scaffold genre sign-flip (reversed at healthy operating points; the original observation came from a marginal-health checkpoint). The hollow-bar rule exists for exactly this.

Prereg	Prediction	Outcome	Verdict
Loop 3 (recipe)	conduct coefficient .95	specific effect $+2$pp over placebo	**MISS**
	rule coefficient .90	-9pp (n=12)	**MISS**
	format coefficient .60	no clean point (precondition failed)	**MISS**
	phrasing / scaffold .50±	51% / 42% raw	HIT / HIT (raw)
	drills ≈ 0	direction right, harm unpredicted	dir. HIT
	A2 arm (measured ratios)	inputs falsified in 4/5 buckets	non-executable
Batch 2 (dose)	all format doses pass gates; saturate by 20%	all pass; saturates at $\leq$10%	HIT*
	drills harm monotone ($\geq$5pp at 10%)	flat on plain items; genre-gated	**MISS**
	scafffmt separates structure vs JSON	falls in JSON interval (attribution stayed)	HIT (rule)
Loop 5 (ΔW)	$\cos(d, \Delta W_{\text{drills}}) < 0$	$+0.13$: decorrelated, not opposed	**MISS**
	single-component ΔW angles	drills = global outlier (all pairwise min)	registered
Tier 2 (frontier)	lookup tier A $\approx$ 0; depth decay	0.2/0.0%; monotone but no bend $\leq$3 steps	5/5 HIT
	M calibration formula	self-neutralised at ceiling (no range)	instrument note
Bendpoint (G/B)	5-step teachable (OOD $\geq$85%)	G<B; OOD .83, overtrain dip from .91	HIT (edge)
	branching = bend candidate	100%/100% — frontier unbent	**MISS**
	NL wrapping costs $\geq$20pp	100%/100% — wrapping is free	**MISS**
E arm (steering)	conduct bucket $\approx$94%	68% $<$ placebo (resist-vs-correct mismatch)	**MISS**
	repair genre $<+10$pp	$+42.7$pp, above all trained arms	**MISS***
M2/M3 (cross-model)	Llama: floor + format-component replicate	both replicate (format = loss-prevention)	2 HIT
	Llama: drills toxicity replicates	absent at 25% dose	**MISS**
	Llama: steering direction transfers	global degradation; CL-3 scoped to Qwen	**MISS**
C-15 (foundations)	D / drills replicate under new seeds	both replicate (D .710±.009)	2 HIT
	scaffold sign-flip replicates	2/3 seeds reverse at healthy ridge	**REVERSE**
	Llama lever at deeper layers; plain cost $\leq$3pp	$+17.3$ / flat / -3.03pp (9 items)	2 HIT + **MISS**
E1 (data size)	targeted $N \leq 300$ suffices; random unstable	mute-wave artifact; measure replaced	**hard stop**

Table 2: Preregistration scorecard. *as with format saturation, the preregistration undershot in the direction of the cheaper method — the miss direction is itself information; the E-arm result passed a four-point hard-stop autopsy before entering any figure (Appendix B). The Loop-3 misses motivated the placebo-delta reading (COR-1); the dose miss became the genre-gating result (§5); the ΔW miss forced the compliance-teaching reframing (COR-3).

Abstain variance warning. The retrieve-or-abstain policy is the highest seed-variance column in the study (uniform arm across three seeds: .34/.15/.45); a single-seed ordering anomaly (D $>$ B $>$ A1) collapsed entirely into this band and is recorded as noise, not finding.

B Instruments and audits

Judge definitions (strict/loose rescue readings, the non-dict guard on the abstain judge), the construct-audit protocol ($\geq$70% acceptance per probe type on 100 manually audited items), pool filters (330 items excluded: negative or non-integer golds), and per-model instrument gates are collected here. The labile-core discount used in Figure 1's footnote derives from Figure 8.

C Retired results, with autopsies

A negative result enters this appendix only with an autopsy: original observation, replication outcome, and the most parsimonious diagnosis together with the rule in the body it exemplifies. Phenomena without an explanation stay in the scorecard.

Scaffold genre sign-flip. *Observation:* on one seed the scaffold component rescued the plain-genre format bucket strongly while scoring 11.9pp below placebo in the repair genre — an apparent sign flip, briefly a third genre-gating case. *Replication:* under two fresh seeds at healthy operating points (epoch 4; the original checkpoint passed gates marginally, with 11% mute, recorded at

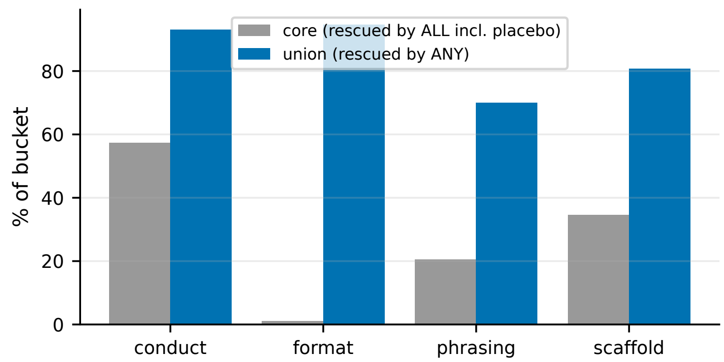

Figure 8: Flip-overlap audit: rescue is not dice. Every bucket has a deterministic core rescued by all arms including the training placebo (conduct 57%); format's near-zero core (1%) is the converse evidence. Observed Jaccard exceeds the independent-rescue baseline in all pairs.

the time), two of three seeds place the repair-genre score *above* placebo, and the plain-genre rescue shrinks from 94% to 41–49%. *Diagnosis:* an operating-point artifact — readings taken at a marginal-health checkpoint are not interpretable, which is precisely the case for the operating-point rule of §2.3. Three seeds cannot fully exclude a seed-fragile real effect; the artifact reading is the most parsimonious.

Abstain-policy ordering. *Observation:* a single-seed ordering of arms on the retrieve-or-abstain policy (D $>$ B $>$ A1) suggested mixture composition drove abstention quality. *Replication:* the uniform arm's abstain score across three seeds spans .15–.45. *Diagnosis:* abstention is the highest seed-variance column in the study; any single-seed ordering within a $\pm$15pp band is noise, which is why every ordering claim in the body is multi-seed. Both autopsies are the hollow-bar rule doing its job.

D Three-seed aggregates for the 2Wiki cell

	adopt (plain)	derail (plain)	repair genre (strict)
pre-repair	19	28	.638
U-mix (3 seeds)	0/1/0	4/11/5	.479/.471/.479
clean replay (3 seeds)	7/7/8	56/42/49	.441/.439/.431

Table 3: 2Wiki three-seed aggregates behind Figure 5: the plain-face separation (anti-credulity transfers) and the repair-genre non-separation are both stable across seeds.

E Ratio-arm statistics

Item-level paired bootstrap (10,000 resamples) over the 480 repair-genre items, arms averaged over three seeds each: Δ(B$-$A1) $= +4.2$pp, 95% CI $[+1.7, +6.7]$; Δ(D$-$A1) $= +3.4$pp $[+0.1, +6.7]$; Δ(B$-$D) $= +0.8$pp $[-2.0, +3.5]$. TOST equivalence at a $\pm$3pp margin (chosen post hoc, at the scale of seed-level standard deviations) is not established for any pair. Prevalence-matched ratios did not outperform uniform or reversed coverage at the tested budgets.

F Claim truth table

The machine-readable master is `paper/CLAIM_TRUTH_TABLE.md` in the repository; every body claim, its model$\times$domain cells, seed counts, evidence files and verdict (replicated / variant / absent / untested / retired) are enumerated there, and every caption in this paper was audited against it before submission. Retired rows carry autopsies (Appendix C).

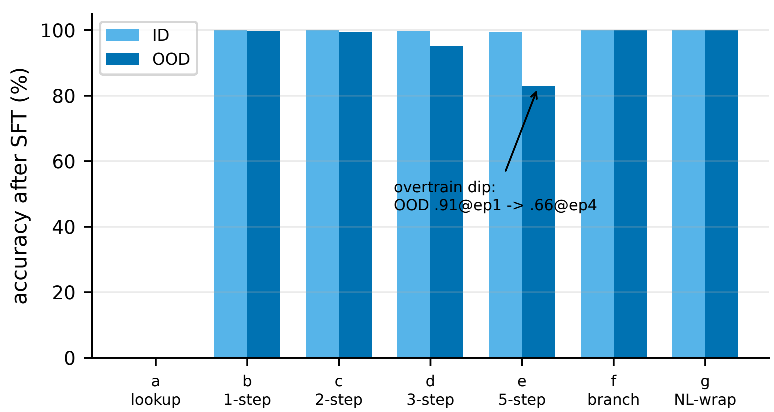

Figure 9: The synthetic learnability frontier (moved from the body). In this synthetic setting, depth, branching and wrapping alone do not create a learning boundary; the claim is scoped to invented-operator tasks.